

XML Code Documentation Report

For task 7, XML comments were added in the code files of the banking application. The reason was to make the code more clear and more easy to understand. These comments explain what the class does, what the method does, and what some inputs and outputs mean. This helps when reading the program later and also helps in maintainance.

The XML comments were written inside the actual C# source files. They were added above important classes, constructors, fields, and methods. After that, XML documentation was enabled in the project build settings in Visual Studio. Then the solution was built again so the XML documentation file could be generated by the system.

Files Documented

The XML comments were added in the main program files. These files included the classes for customer data, account data, controller logic, persistence logic, and form code.

The main documented files were:

- BankController.cs
- Customer.cs
- Account.cs
- EverydayAccount.cs
- InvestmentAccount.cs
- OmniAccount.cs
- MainForm.cs
- ManageAccountsForm.cs
- BankData.cs
- CustomerData.cs
- AccountData.cs

These files were chosen because they hold the main logic and structure of the banking prototype.

Documentation Process

The documentation process was done in a few steps. First, XML comments were inserted into the source code. These comments used summary tags and method details. Second, the project

settings were updated so XML documentation output could be generated. Third, the project was built again and the XML file was created in the output folder.

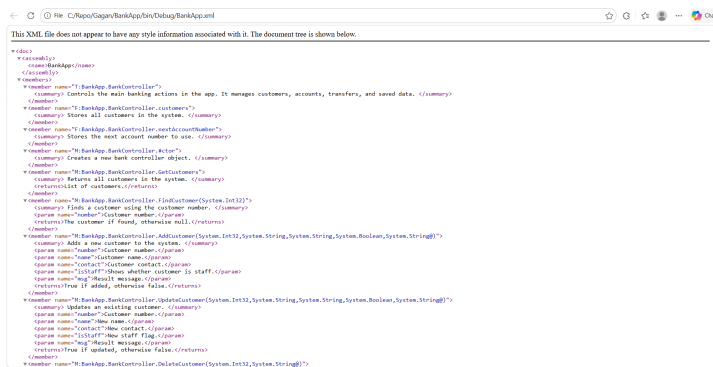
This process showed that the comments were not just typed in the code, but were also compiled into a proper documentation file. This made the task more complete and more correct.

Documentation Output

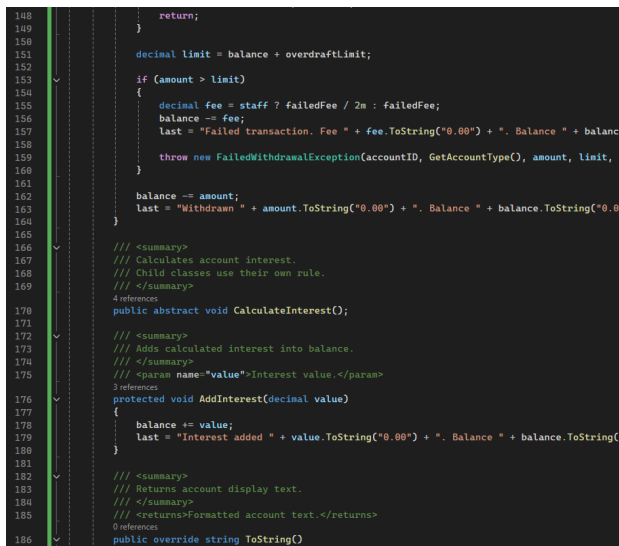
After rebuilding the project, the XML documentation file was generated successfully. This file confirmed that the XML comments were recognised by the system and converted into a documentation output file.

The output file generated was:

BankApp.xml



Screenshot Evidence



```

namespace BankApp
{
    /// <summary>
    /// Represents an everyday account.
    /// This account has no interest and no overdraft.
    /// </summary>
    3 references
    public class EverydayAccount : Account
    {
        /// <summary>
        /// Creates a new everyday account.
        /// </summary>
        /// <param name="id">Account id.</param>
        /// <param name="opening">Opening balance.</param>
        2 references
        public EverydayAccount(int id, decimal opening)
        {
            : base(id, opening, 0m, 0m, 0m)
        }

        /// <summary>
        /// Everyday account does not earn interest.
        /// </summary>
        2 references
        public override void CalculateInterest()
        {
            AddInterest(0m);
        }
    }
}

```

```

22
23 //<summary>
24 //<summary> Calculates interest for investment account.
25 //<summary> Interest applies when balance is above 1000.
26 //</summary>
27 // 2 references:
28 public override void CalculateInterest()
29 {
30     decimal interest = 0m;
31     //<summary>
32     //<summary> if (balance > 1000m)
33     {
34         interest = balance * (interestRate / 100m);
35     }
36     AddInterest(interest);
37 }
38
39 //<summary>
40 //<summary> Withdraws money from investment account.
41 //<summary> This account cannot go below available balance.
42 //</summary>
43 //<param name="amount">Withdrawal amount.</param>
44 //<param name="staff">Shows whether customer is staff.</param>
45 // 5 references:
46 public override void Withdraw(decimal amount, bool staff)
47 {
48     if (amount <= 0)
49     {
50         last = "Amount must be positive.";
51         return;
52     }
53     if (amount > balance)
54     {
55         decimal fee = staff ? failedFee / 2m : failedFee;
56         balance -= fee;
57         last = "Failed transaction. Fee " + fee.ToString("0.00") + ". Balance " + balance;
58         throw new FailedWithdrawalException(accountID, GetAccountType(), amount, balance);
59     }

```